

# Atomistic DWBA for Unit Cells in a Graded Interface

orGUI design note

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## Abstract

This note specifies the first-order distorted-wave Born approximation needed to add finite atomic models to orGUI's graded optical reference. A generated `UnitCell`, whether it belongs to a `Film`, an `EpitaxyInterface`, or a surface model, is evaluated in every optical medium that contains its atoms. Its amplitude is the coherent sum of the four Renaud incident/exit channels. For non-specular scattering the planar reference has no Fourier component and the atomic sum is the complete contrast amplitude. For specular scattering the constant  $\delta_m + i\beta_m$  density of every prepared optical slab remains in the matrix element and must be subtracted analytically. The subtraction is partitioned among unit cells without changing the profile-level result, so signed occupancies, strain, lateral-area conversion, and optical-profile simplification remain consistent.

# Contents

|          |                                                                 |          |
|----------|-----------------------------------------------------------------|----------|
| <b>1</b> | <b>Reference problem and conventions</b>                        | <b>1</b> |
| 1.1      | Scope . . . . .                                                 | 1        |
| 1.2      | Fourier, surface, and refractive-index conventions . . . . .    | 1        |
| 1.3      | Scattering-factor density and the DWBA matrix element . . . . . | 2        |
| <b>2</b> | <b>Atomic unit cells and the four coherent channels</b>         | <b>3</b> |
| 2.1      | Prepared field channels in one optical medium . . . . .         | 3        |
| 2.2      | One generated UnitCell record . . . . .                         | 4        |
| 2.3      | Useful limits . . . . .                                         | 4        |
| <b>3</b> | <b>Non-specular and specular contrast amplitudes</b>            | <b>5</b> |
| 3.1      | Why the two cases differ . . . . .                              | 5        |
| 3.2      | Integral of one constant optical slab . . . . .                 | 5        |
| 3.3      | A well-defined per-unit-cell subtraction . . . . .              | 6        |
| 3.4      | Reduction to the implemented terminal bulk formula . . . . .    | 7        |
| <b>4</b> | <b>Mapping to orGUI models and validation</b>                   | <b>8</b> |
| 4.1      | Flattened atomic representation . . . . .                       | 8        |
| 4.2      | Preparation and evaluation boundary . . . . .                   | 8        |
| 4.3      | Specular observable . . . . .                                   | 9        |
| 4.4      | Regression invariants . . . . .                                 | 9        |
| 4.5      | References . . . . .                                            | 10       |

# Chapter 1

## Reference problem and conventions

### 1.1 Scope

The unperturbed electric fields and the unperturbed specular reflection amplitude are assumed to be available from the layered-field solver. The problem addressed here is the remaining atomistic matrix element for finite unit cells in the graded interface region. The terminal semi-infinite bulk is included only to show that the new formula reduces to the already implemented bulk result.

The derivation is electromagnetic throughout. Following Renaud, Lazzari, and Leroy, the incident field and the reciprocal field launched from the detector are denoted by electric fields  $\mathbf{E}_0$ ; no quantum-mechanical wavefunction notation is used. The medium is nonmagnetic, isotropic, linear, and reciprocal. Magnetic and tensor-resonant scattering are outside the present scope.

### 1.2 Fourier, surface, and refractive-index conventions

We use the crystallographic space-time convention

$$\mathbf{E}(\mathbf{r}, t) \propto \exp[i(\omega t - \mathbf{k} \cdot \mathbf{r})], \quad k_0 = \frac{2\pi}{\lambda}, \quad \mathbf{Q} = \mathbf{k}_f - \mathbf{k}_i, \quad (1.1)$$

and the Fourier transform

$$\tilde{g}(\mathbf{Q}) = \int g(\mathbf{r}) \exp(i\mathbf{Q} \cdot \mathbf{r}) \, d^3r. \quad (1.2)$$

The  $z$  axis is the outward surface normal: vacuum is at large positive  $z$  and the bulk extends toward negative  $z$ . Momentum transfer is in  $\text{\AA}^{-1}$  and all real-space coordinates below are in  $\text{\AA}$ .

The prepared optical reference consists of planar media  $m = 0, \dots, M$ . In medium  $m$ ,

$$n_m = 1 - \delta_m - i\beta_m, \quad \delta_m, \beta_m \in \mathbb{R}, \quad (1.3)$$

and the medium occupies the interval

$$I_m = (z_m^-, z_m^+). \quad (1.4)$$

One or both endpoints may be infinite for a terminal medium. A fixed half-open convention must be used in code when an atom lies exactly on a boundary.

orGUI uses the negative-root Renaud normal wavevector

$$k_{z,m} = -k_0 \sqrt{n_m^2 - n_0^2 \cos^2 \alpha}, \quad \text{Re } k_{z,m} \leq 0, \quad \text{Im } k_{z,m} \geq 0. \quad (1.5)$$

The incident and reciprocal exit solutions are calculated in the same frozen array of complex refractive indices and at the same physical boundaries.

### 1.3 Scattering-factor density and the DWBA matrix element

For atom  $a$ , orGUI uses

$$f_a(Q^2, E) = f_{a,\text{WK}}^0(Q^2) + f'_a(E) + \text{i}f''_a(E), \quad (1.6)$$

where the Waasmaier–Kirfel representation is evaluated analytically at the generally complex channel value  $Q^2$ . Its zero-transfer value, including the same anomalous terms, is also used to generate the optical profile. Replacing that value by an independent atomic-number approximation would break the reference-density cancellation.

Let  $\rho_f(\mathbf{r})$  be the complex scattering-factor density in electrons per Angstrom<sup>3</sup>. To first order in the X-ray susceptibility,

$$n^2(\mathbf{r}) \simeq 1 - \frac{4\pi r_e}{k_0^2} \rho_f(\mathbf{r}), \quad (1.7)$$

with  $r_e$  the classical electron radius. The piecewise-constant reference density represented by Eq. (1.3) is therefore

$$\boxed{\bar{\rho}_{f,m} = \frac{k_0^2}{2\pi r_e} (\delta_m + \text{i}\beta_m).} \quad (1.8)$$

Define

$$\Delta\rho_f(\mathbf{r}) = \rho_f(\mathbf{r}) - \sum_m \bar{\rho}_{f,m} \mathbf{1}_{I_m}(z). \quad (1.9)$$

The first-order far-field correction for an analyzed exit polarization is

$$\delta\mathbf{E}_f(\mathbf{r}) = -r_e \frac{\text{e}^{-\text{i}k_0 r}}{r} \boldsymbol{\epsilon}_f F_{\Delta,\text{DWBA}}, \quad (1.10)$$

where the electron-unit contrast matrix element is

$$\boxed{F_{\Delta,\text{DWBA}} = \int \mathbf{E}_0(\mathbf{r}, -\mathbf{k}_f) \cdot \mathbf{E}_0(\mathbf{r}, \mathbf{k}_i) \Delta\rho_f(\mathbf{r}) \, \text{d}^3r.} \quad (1.11)$$

The reciprocal exit field is not complex conjugated. Complex conjugation enters only after the coherent amplitude has been formed, for example in  $r_e^2 |F_{\Delta,\text{DWBA}}|^2$ .

Equation (1.11) is the only starting point needed for both the atomic and constant-profile terms. Reflection, transmission, refraction, and repeated reflection at the planar interfaces are already contained in the two reference electric fields.

## Chapter 2

# Atomic unit cells and the four coherent channels

### 2.1 Prepared field channels in one optical medium

Every finite optical medium contains a downward and an upward branch of both the incident and reciprocal exit electric fields. Let

$$\sigma_i, \sigma_f \in \{+1, -1\}, \quad \nu = (\sigma_i, \sigma_f), \quad (2.1)$$

with the fixed channel order

$$++, \quad +-, \quad -+, \quad --. \quad (2.2)$$

For medium  $m$ , channel  $\nu$  has

$$\mathbf{Q}_m^\nu = (\mathbf{Q}_\parallel, Q_{z,m}^\nu), \quad Q_{z,m}^\nu = -(\sigma_i k_{z,m,i} + \sigma_f k_{z,m,f}), \quad (2.3)$$

and

$$(Q_m^\nu)^2 = |\mathbf{Q}_\parallel|^2 + (Q_{z,m}^\nu)^2. \quad (2.4)$$

Neither  $Q_{z,m}^\nu$  nor  $(Q_m^\nu)^2$  is generally real in an absorbing medium.

Let  $A_{m,i}^{\sigma_i}$  and  $A_{m,f}^{\sigma_f}$  be the local Renaud electric- field amplitudes, referenced by the field solver to  $z_{m,i}^{\text{ref}}$  and  $z_{m,f}^{\text{ref}}$ . The complete coefficient multiplying a Fourier factor  $\exp(i\mathbf{Q}_m^\nu \cdot \mathbf{r})$  is

$$\begin{aligned} \mathcal{C}_{m,\nu} &= A_{m,i}^{\sigma_i} A_{m,f}^{\sigma_f} P_{m,\nu} \\ &\times \exp\left\{i \left[ \sigma_i k_{z,m,i} z_{m,i}^{\text{ref}} + \sigma_f k_{z,m,f} z_{m,f}^{\text{ref}} \right]\right\}. \end{aligned} \quad (2.5)$$

The reference phase in Eq. (2.5) is not optional: the stored  $A^\pm$  are local amplitudes, not amplitudes referenced to the global crystallographic origin.

For the right-handed Renaud polarization basis used by orGUI, the bilinear polarization contractions are

$$\begin{aligned} P_{ss}^{\sigma_i \sigma_f} &= \cos \varphi, \\ P_{sp}^{\sigma_i \sigma_f} &= -\sigma_f \sin \alpha_f \sin \varphi, \\ P_{ps}^{\sigma_i \sigma_f} &= -\sigma_i \sin \alpha_i \sin \varphi, \\ P_{pp}^{\sigma_i \sigma_f} &= \sin \alpha_i \sin \alpha_f \left[ \frac{k_0^2 \cos \alpha_i \cos \alpha_f}{k_{z,m,i} k_{z,m,f}} - \sigma_i \sigma_f \cos \varphi \right]. \end{aligned} \quad (2.6)$$

The solver's upward internal tangential coefficient for  $p$  polarization has the opposite sign from the public Renaud amplitude. Thus

$$A_{p,m}^+ = T_{p,m}^+, \quad A_{p,m}^- = -T_{p,m}^-. \quad (2.7)$$

Implementations should use the solver's analytic  $A^\sigma/k_z$  limits in the  $pp$  expression rather than divide numerically at a critical angle.

## 2.2 One generated UnitCell record

The surface model is flattened into generated unit-cell records  $u$ . Before flattening, every source **UnitCell** is split into the same ordered structural-layer views used to build the optical profile. A record therefore contains one such layer cell, a crystal-model weight  $w_u$ , its coherent domain transforms  $d$ , and possibly signed domain occupancies  $p_{ud}$ . A one-layer source cell is already a single record. Let  $A_u$  be the source cell's lateral area and  $A_{\text{ref}}$  the crystal-wide reference area. The required area conversion is

$$s_u = \frac{A_{\text{ref}}}{A_u}. \quad (2.8)$$

Current Film and interface domain transforms preserve lateral area. If a future model permits in-plane strain,  $A_u$  in Eq. (2.8) must be replaced by the transformed projected area, including its Jacobian.

After stacking and the coherent-domain transform, atom  $a$  of record  $u$  and domain  $d$  is at the physical Cartesian position  $\mathbf{r}_{uad}$ . The atom is assigned to the unique prepared optical interval

$$m(u, a, d) \quad \text{such that} \quad z_{uad} \in I_{m(u,a,d)}. \quad (2.9)$$

A generated unit cell that crosses an optical boundary must therefore be split by atom. Assigning one medium to the entire cell would use the wrong  $A^\pm$ ,  $k_z$ , polarization contraction, and complex form-factor argument for some of its atoms.

For channel  $\nu$  in medium  $m$ , define the atomic contribution

$$\mathcal{A}_{u,m}^\nu = s_u w_u \mathcal{C}_{m,\nu} \sum_d p_{ud} \sum_{\substack{a \in u \\ m(u,a,d)=m}} o_a f_a((Q_m^\nu)^2, E) \exp[-W_a(Q_m^\nu)] \exp(i\mathbf{Q}_m^\nu \cdot \mathbf{r}_{uad}). \quad (2.10)$$

Here  $o_a$  is the atomic occupancy. For the currently implemented in-plane and out-of-plane Debye-Waller parameters,

$$W_a(Q_m^\nu) = \frac{B_{a,\parallel} |\mathbf{Q}_m^\nu|^2 + B_{a,\perp} (Q_{z,m}^\nu)^2}{16\pi^2}. \quad (2.11)$$

Equations (1.6), (2.4), and (2.11) are evaluated at complex arguments without taking an absolute value or a real part.

The complete atomic amplitude of record  $u$  is

$$\boxed{F_{\text{atomic},u} = \sum_m \sum_{\nu \in \{++,+,-,-\}} \mathcal{A}_{u,m}^\nu}. \quad (2.12)$$

The four terms in Eq. (2.12) are amplitudes and must be added before an intensity is formed. The incorrect expression  $\sum_\nu |\mathcal{A}_{u,m}^\nu|^2$  discards the standing-wave interference between incident and exit branches.

## 2.3 Useful limits

In a terminal substrate, physical boundary conditions give

$$A_{M,i}^- = A_{M,f}^- = 0, \quad (2.13)$$

so only the  $++$  channel survives. In the high-angle Born limit of a finite cell in vacuum,  $A_i^+ = A_f^+ = 1$  and all reflected branches vanish; the same reduction occurs. These are limiting cases of the four-channel expression, not reasons to omit channels in finite graded layers.

If the cell is wholly contained in one medium, Eq. (2.10) reduces to four ordinary structure-factor evaluations at the four complex values  $Q_{z,m}^\nu$ . If it crosses a boundary, there is no single structure factor  $F_u(h, k, l)$  that can be multiplied by one set of channel coefficients; the atom-medium partition in Eq. (2.10) is essential.

## Chapter 3

# Non-specular and specular contrast amplitudes

### 3.1 Why the two cases differ

The reference density is translationally invariant parallel to the surface. Its lateral Fourier transform is

$$\int_{\mathbb{R}^2} \exp(i\mathbf{Q}_{\parallel} \cdot \mathbf{r}_{\parallel}) \, d^2r_{\parallel} = (2\pi)^2 \delta^{(2)}(\mathbf{Q}_{\parallel}). \quad (3.1)$$

Consequently, the constant  $\delta_m, \beta_m$  term is absent at a genuinely non-specular point but remains on the specular rod. This is a statement about the laterally infinite optical reference. Subtracting a rectangular “reference patch” for every unit cell off-specular would replace the Dirac distribution by a finite-cell shape transform and create artificial diffuse scattering.

The two returned matrix elements must therefore be defined separately:

$$F_{\Delta,u}^{\text{nonsp}} = F_{\text{atomic},u}, \quad \mathbf{Q}_{\parallel} \neq \mathbf{0}, \quad (3.2)$$

$$F_{\Delta,u}^{\text{spec}} = F_{\text{atomic},u} - F_{\text{ref},u}, \quad \mathbf{Q}_{\parallel} = \mathbf{0}. \quad (3.3)$$

Instrumental resolution may mix diffuse intensity into a detector pixel on the specular rod, but it does not change Eqs. (3.2) and (3.3). That mixing belongs to a later resolution convolution, not to an interpolation of the reference subtraction near  $\mathbf{Q}_{\parallel} = \mathbf{0}$ .

### 3.2 Integral of one constant optical slab

Within prepared medium  $m$ , both  $\bar{\rho}_{f,m}$  and every channel coefficient are constant; only the propagation phase varies with  $z$ . Define the stable depth integral

$$\Phi(q; a, b) = \int_a^b e^{iqz} \, dz = \begin{cases} \frac{e^{iqb} - e^{iqua}}{iq}, & q \neq 0, \\ b - a, & q = 0. \end{cases} \quad (3.4)$$

For numerical work, an equivalent midpoint form is preferable:

$$\Phi(q; a, b) = (b - a) e^{iq(a+b)/2} \text{sinc}\left(\frac{q(b-a)}{2}\right), \quad \text{sinc } x = \frac{\sin x}{x}, \quad (3.5)$$

or the numerator of Eq. (3.4) may be evaluated with a complex `expm1`. For a lower terminal half-space, with the physical decaying branch,

$$\Phi(q; -\infty, b) = \frac{e^{iqb}}{iq}. \quad (3.6)$$



Normalized to one lateral reference cell, the complete reference term of the prepared profile is

$$F_{\text{ref,DWBA}}^{\text{spec}} = A_{\text{ref}} \sum_m \bar{\rho}_{f,m} \sum_{\nu \in \{++,+,-,-+,-\}} \mathcal{C}_{m,\nu} \Phi(Q_{z,m}^\nu; z_m^-, z_m^+). \quad (3.7)$$

Thus the specular result keeps the integral over the constant  $\delta_m + i\beta_m$  term explicitly. Finite graded layers generally retain all four terms in Eq. (3.7); only the terminal substrate reduces to  $++$ .

### 3.3 A well-defined per-unit-cell subtraction

Only the total reference profile is physical. A decomposition of that profile among overlapping model objects is bookkeeping and is not unique unless a rule is fixed. orGUI can make the decomposition deterministic by retaining the forward-density contribution of every generated record while building the profile.

Let  $\bar{\rho}_{f,u,m}$  denote record  $u$ 's share of the constant reference density in prepared medium  $m$ . The required conservation rule is

$$\sum_u \bar{\rho}_{f,u,m} = \bar{\rho}_{f,m} \quad \text{for every prepared medium } m. \quad (3.8)$$

Before optical-profile simplification, a generated slab of physical thickness  $t_{ud}$  contributes naturally

$$\bar{\rho}_{f,ud} = \frac{w_u p_{ud}}{A_u t_{ud}} \sum_{a \in u} o_a f_a(0, E). \quad (3.9)$$

Normal strain changes  $t_{ud}$  and hence the volume density, but preserves the integrated areal content. Signed occupancies in interface and etching models remain signed in Eq. (3.9).

Coincident profile contributions are added. If adjacent optical media are merged, first determine the merged boundaries from the total profile, then thickness-average every record's density over those same boundaries. This is a linear operation once the grouping is fixed and preserves Eq. (3.8). Reconstructing shares independently with different merge decisions would not preserve the prepared reference.

The per-record reference amplitude is then

$$F_{\text{ref},u} = A_{\text{ref}} \sum_m \bar{\rho}_{f,u,m} \sum_\nu \mathcal{C}_{m,\nu} \Phi(Q_{z,m}^\nu; z_m^-, z_m^+), \quad (3.10)$$

and the requested individual-cell specular contrast is

$$F_{\Delta,u}^{\text{spec}} = \sum_m \sum_\nu \mathcal{A}_{u,m}^\nu - A_{\text{ref}} \sum_m \bar{\rho}_{f,u,m} \sum_\nu \mathcal{C}_{m,\nu} \Phi(Q_{z,m}^\nu; z_m^-, z_m^+). \quad (3.11)$$

Summing Eq. (3.11) over  $u$  and applying Eq. (3.8) recovers the profile-level identity

$$F_{\Delta,\text{DWBA}}^{\text{spec}} = \sum_u F_{\Delta,u}^{\text{spec}} = \sum_u F_{\text{atomic},u} - F_{\text{ref,DWBA}}^{\text{spec}}. \quad (3.12)$$

If callers do not need an attribution by object, Eq. (3.12) is the preferred calculation: accumulate all atomic terms, evaluate the prepared reference once, and subtract once.

### 3.4 Reduction to the implemented terminal bulk formula

Let the top terminal bulk cell occupy  $b - c \leq z < b$ , with repeat vector  $-\mathbf{c}$  into the substrate and  $c = c_z > 0$ . Only the  $++$  channel survives. If  $F_{\text{uc}}^{\text{atomic}}$  includes the global phase of the top cell, its semi-infinite atomic sum is

$$F_{\text{atomic,bulk}} = \mathcal{C}_{M,++} \frac{F_{\text{uc}}^{\text{atomic}}}{1 - \exp(-i\mathbf{Q}_M^{++} \cdot \mathbf{c})}. \quad (3.13)$$

For a surface-normal repeat, the mean-density term is

$$F_{\text{ref,bulk}} = \mathcal{C}_{M,++} A_{\text{ref}} \bar{\rho}_{f,M} \frac{e^{iQ_{z,M}^{++} b}}{iQ_{z,M}^{++}}. \quad (3.14)$$

Equations (3.13) and (3.14) are the previous bulk result. The four-channel finite-layer formula therefore extends the existing kernel without changing the terminal half-space convention.

## Chapter 4

# Mapping to orGUI models and validation

### 4.1 Flattened atomic representation

The mathematical kernel should consume a flattened list of generated atomic records rather than special-case each high-level model. After stacking, the records have the following meanings.

| Model                         | Records supplied to the DWBA sum                                                                                                                                                                                                                 |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>UnitCell</code>         | Ordered structural-layer views, atoms, source lateral area, internal coherent-domain transforms and occupancies, and the crystal-wide component weight.                                                                                          |
| <code>Film</code>             | The finite cyclic layer cells generated by <code>createLayers</code> . Their domain translations place them at the physical stacked heights.                                                                                                     |
| <code>EpitaxyInterface</code> | Positive upper-material probability records; positive lower-material records at their possibly strain-coupled positions; and negative lower-material records that remove the sharp, unstrained bulk already present in the terminal lattice sum. |
| <code>PoissonSurface</code>   | Positive records for grown material and negative records for material removed from the sharp Film termination. Termination corrections remain coherent with the Film amplitude.                                                                  |

Negative occupancies are amplitudes, not negative intensities. They must pass unchanged through Eqs. (2.10) and (3.9). Likewise, alternative crystal domains are applied as coherent amplitudes within each configured domain; intensities are combined only at the model level where the existing API explicitly defines an incoherent mixture.

Water and other continuous-density models do not have an atomic `UnitCell` sum and are outside this chapter. They may use the same slab integral in Eq. (3.7), but their actual-density integral must be derived from their own continuous profile.

### 4.2 Preparation and evaluation boundary

The prepared object must freeze all quantities that define the reference problem:

- the physical optical boundaries and the final, possibly simplified, values  $\delta_m, \beta_m$ ;
- incident and reciprocal exit  $k_z$ ,  $A^\pm$ ,  $A^\pm/k_z$ , and reference planes for every medium;
- incident and exit polarization, external geometry, and  $Q_\parallel$ ;

- the reference-area convention and the per-record reference shares satisfying Eq. (3.8).

Atomic positions, occupancies, Debye–Waller parameters, and species may be reevaluated against a frozen reference if that is the intended fit operation. Changing lattice transforms, stacking geometry, photon energy, optical boundaries, profile simplification, or the reference area invalidates the preparation. Rebuilding only the atomic list while silently changing the optical profile would combine two different reference problems.

For each scattering point, evaluation is naturally ordered as follows:

1. calculate  $\mathbf{Q}_m^\nu$ ,  $P_{m,\nu}$ , and  $\mathcal{C}_{m,\nu}$  for all media and all four channels;
2. assign every transformed atom to its physical optical medium;
3. accumulate Eq. (2.10) coherently over atoms, domains, media, and channels;
4. if the point is specular, evaluate the constant-profile integral in Eq. (3.7) and subtract it; otherwise leave the atomic amplitude unchanged;
5. only then form an intensity or combine the scattered specular field with the prepared unperturbed reflection amplitude.

### 4.3 Specular observable

For a finite contrast normalized to one reference lateral cell, the scattered far-field kernel is  $r_e^2 |F_{\Delta, \text{DWBA}}|^2$ . On the ideal specular rod the result is instead combined coherently with the prepared plane-wave reflection amplitude. Using the same normalization as the existing bulk derivation,

$$\delta r_{\text{DWBA}} = -\frac{2\pi i r_e}{A_{\text{ref}} k_{z,0,f}} F_{\Delta, \text{DWBA}}^{\text{spec}}, \quad (4.1)$$

and

$$r_{\text{total}} = r_0 + \delta r_{\text{DWBA}}, \quad R_{\text{coh}} = |r_{\text{total}}|^2. \quad (4.2)$$

The  $r_0$  in Eq. (4.2) must come from the same prepared incident field as the  $A_{m,i}^\pm$  used in the matrix element. For unpolarized radiation, calculate  $s$  and  $p$  independently and average their intensities; do not add their amplitudes.

### 4.4 Regression invariants

The following checks are required before the atomistic extension is considered scientifically usable.

1. **Four-channel coherence.** A finite layer with nonzero  $A_i^\pm$  and  $A_f^\pm$  equals the direct atom-by-atom evaluation of Eq. (1.11); changing to a sum of channel intensities must fail the test.
2. **Born limit.** Setting  $A_i^+ = A_f^+ = 1$  and the other branches to zero recovers the conventional UnitCell structure factor and the standard polarization contraction.
3. **Terminal limit.** A finite stack made progressively thicker converges to Eqs. (3.13) and (3.14) away from exact Bragg poles.
4. **Identical actual and reference density.** Replacing every optical slab by its exactly matching atomic forward density gives  $F_{\Delta, \text{DWBA}}^{\text{spec}} = 0$  within the atomic-truncation accuracy.
5. **Off-specular invariance.** Adding the reference-density bookkeeping does not change any result with  $\mathbf{Q}_\parallel \neq \mathbf{0}$ .

6. **Partition invariance.** Summing per-record results from Eq. (3.11) equals the single profile-level subtraction in Eq. (3.12), including after profile coalescing and simplification.
7. **Area invariance.** Two commensurate descriptions with different source lateral cells give the same amplitude per  $A_{\text{ref}}$  after Eq. (2.8) is applied to both atomic and continuum terms.
8. **Signed-model identity.** At zero width, zero growth, or zero etching, the positive and negative interface/surface records cancel in both the atomic and reference-profile sums.
9. **Boundary split.** Moving an atom across an optical boundary changes its channel coefficient and  $Q_z$  exactly once; a cell straddling a boundary agrees with the same atom list represented as two records.
10. **Low- $Q_z$  stability.** Equation (3.4) approaches the slab thickness without catastrophic cancellation.
11. **Reciprocity.** Exchanging source and detector geometry obeys the reciprocal-field symmetry without conjugating the exit field.
12. **No duplicate attenuation.** Absorption carried by complex  $Q_z$  is not also applied as an empirical depth factor.

## 4.5 References

The electric-field, signed- $k_z$ , reciprocal-exit-field, and four-channel semantics follow G. Renaud, R. Lazzari, and F. Leroy, “Probing surface and interface morphology with grazing incidence small angle X-ray scattering,” *Surface Science Reports* **64** (2009), 255–380, especially Sections 5.2–5.4, Eq. (91), and the four-event finite-layer examples in Section 6.2. The present note retains their electromagnetic derivation but does not adopt the scalar wavefunction notation used for the optional quantum- mechanical analogy.

The terminal crystal-truncation result and the atomic-density truncation caveat are cross-checked against V. M. Kaganer, “Crystal truncation rods in kinematical and dynamical x-ray diffraction theories,” *Physical Review B* **75** (2007), 245425. The perturbation as a difference from a planar reference follows the same physical construction used by G. H. Vineyard, *Physical Review B* **26** (1982), 4146–4159, and by S. K. Sinha *et al.*, *Physical Review B* **38** (1988), 2297–2311.

The companion file `dwba_literature_crosscheck.md` records the broader comparison with the surface-DWBA and reflectivity literature.